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Q1. String Character Frequency Counter

CODE

```
s = input().lower()
d = {}
for i in s:
    if i != " ":
        d[i] = d.get(i,0)+1
for i in sorted(d):
    print (i, "→", d[i])
```

INPUT

Hello World

OUTPUT

```
d -> 1
e -> 1
h -> 1
l -> 3
o -> 2
r -> 1
w -> 1
```

Q2. Vowel and Consonant Classifier with Statistics

CODE

```
s = input().lower()
v=0
c=0
vd={}
cd={}
for ch in s:
    if ch in "aeiou":
        v+=1
        vd[ch]=vd.get(ch,0)+1
    elif ch.isalpha():
        c+=1
        cd[ch]=cd.get(ch,0)+1
print ("vowels:",v)
print ("consonents:",c)
if vd:
    a = max(vd, key=vd.get)
    print ("most frequent vowel:",a,vd[a])
if cd:
    b = min(cd,key=cd.get)
    print ("least frequent consonant:",b,cd[b])
print ("ratio",round(v/c,2) if c else 0)
```

INPUT

python programming

OUTPUT

```
vowels: 4
consonents: 13
most frequent vowel: o 2
least frequent consonant: y 1
ratio 0.31
```

Q3. String Compression and Run-Length Encoding

CODE

```
s = input()
r = ""
c = 1
for i in range(len(s)):
    if i<len(s)-1 and s[i] == s[i+1]:
        c+=1
    else:
        r+=s[i]+(str(c)if c>1 else "")
        c = 1
print ("Compressed:", r if len(r)<len(s)else s)
```

INPUT

aabbbccdddee

OUTPUT

Compressed: a2b3c2d4e2

Q4. Advanced Text Statistics Analyser

CODE

```
s = input()
words = s.split()
sent = s.count(".") + s.count("!") + s.count("?")
print ("words:", len(words))
print ("sentences:", sent)
print ("paragraphs:", 1)
avg = sum(len(i) for i in words) / len(words)
print ("avg word length:", round(avg, 2))
if sent:
    print ("avg sentence length:", round(len(words) / sent, 2))
```

INPUT

python is a programming language

OUTPUT

```
words: 5
sentences: 0
paragraphs: 1
avg word length: 5.6
```

Student 8 **Guvvala Praneeth Kumar Reddy** 192525197RC

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Q1. String Character Frequency Counter

— Not Submitted —

Q2. Vowel and Consonant Classifier with Statistics

— Not Submitted —

Q3. String Compression and Run-Length Encoding

— Not Submitted —

Q4. Advanced Text Statistics Analyser

— Not Submitted —